Data Warehouse Distribué avec Apache Hive

Partie 1 – Mise en place de l’environnement (Docker)

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker-compose up -d
[+] up 5/5
 ✓ Network hive           Created
 ✓ Container postgres     Started
 ✓ Container metastore    Started
 ✓ Container hiveserver2  Started
 ✓ Container superset     Started
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive>
```

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it hiveserver2 bash
root@e97a8b94daa5:/opt/hive#
```

```
root@e97a8b94daa5:/opt/hive# beeline -u jdbc:hive2://localhost:10000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://localhost:10000
Connected to: Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
Beeline version 4.0.0 by Apache Hive
0: jdbc:hive2://localhost:10000>
```

Partie 2 – Conception d’un entrepôt de données (schéma en étoile)

```
0: jdbc:hive2://localhost:10000> CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRING, pays STRING );
INFO  : Compiling command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b): CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRI
NG, pays STRING )
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO  : Completed compiling command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b); Time taken: 1.436 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b): CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRI
NG, pays STRING )
INFO  : Starting task [Stage-0:DDL] in serial mode
ERROR : Failed
org.apache.hadoop.hive.ql.metadata.HiveException: AlreadyExistsException(message:Table hive.default.dim_client already exists)
        at org.apache.hadoop.hive.ql.metadata.Hive.createTable(Hive.java:1380) ~[hive-exec-4.0.0.jar:4.0.0]
        at org.apache.hadoop.hive.ql.metadata.Hive.createTable(Hive.java:1388) ~[hive-exec-4.0.0.jar:4.0.0]
        at org.apache.hadoop.hive.ql.ddl.table.create.CreateTableOperation.createTableNonReplaceMode(CreateTableOperation.java:158) ~[hive-exec-4.0.0.jar:4.
0.0]
```

```
0: jdbc:hive2://localhost:10000> CREATE TABLE dim_produit (   id_produit INT,   nom_produit STRING,   categorie STRING );
INFO  : Compiling command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc): CREATE TABLE dim_produit (   id_produit INT,   nom_produit STRI
NG,   categorie STRING )
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO  : Completed compiling command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc); Time taken: 0.01 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc): CREATE TABLE dim_produit (   id_produit INT,   nom_produit STRI
NG,   categorie STRING )
INFO  : Starting task [Stage-0:DDL] in serial mode
ERROR : Failed
```

```
ERROR : FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.dim_temps a
lready exists)
INFO  : Completed executing command(queryId=hive_20260329155027_c4e9c51b-0dc8-4796-b0d7-f228d5d038ef); Time taken: 0.034 seconds
Error: Error while compiling statement: FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(messag
e:Table hive.default.dim_temps already exists) (state=08S01,code=40000)
0: jdbc:hive2://localhost:10000>
```

```
ERROR : FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.ventes already exis
ts)
INFO  : Completed executing command(queryId=hive_20260329155152_dbfd3f57-3e6b-4d2c-9f13-0c5a2aa91bed); Time taken: 0.034 seconds
Error: Error while compiling statement: FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table
hive.default.ventes already exists) (state=08S01,code=40000)
0: jdbc:hive2://localhost:10000>
```

Partie 3 – Chargement des données dans Hive

```
0: jdbc:hive2://localhost:10000> SELECT * FROM ventes;
INFO  : Compiling command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261): SELECT * FROM ventes
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:ventes.id_vente, type:int, comment:null), FieldSchema(name:ventes.id_client, type:int, comment:nu
ll), FieldSchema(name:ventes.id_produit, type:int, comment:null), FieldSchema(name:ventes.id_temps, type:int, comment:null), FieldSchema(name:ventes.montant, type:f
loat, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261); Time taken: 2.903 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261): SELECT * FROM ventes
INFO  : Completed executing command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261); Time taken: 0.0 seconds
+-----------------+-------------------+--------------------+------------------+-----------------+
| ventes.id_vente | ventes.id_client  | ventes.id_produit  | ventes.id_temps  | ventes.montant  |
+-----------------+-------------------+--------------------+------------------+-----------------+
| 1               | 1                 | 101                | 202301           | 150.5           |
| 2               | 2                 | 102                | 202301           | 200.0           |
| 3               | 3                 | 103                | 202301           | 120.75          |
| 4               | 1                 | 104                | 202302           | 180.25          |
| 5               | 4                 | 101                | 202303           | 220.3           |
| 6               | 5                 | 102                | 202303           | 210.9           |
| 7               | 2                 | 103                | 202304           | 185.6           |
| 8               | 3                 | 104                | 202304           | 160.8           |
| 9               | 1                 | 101                | 202305           | 145.3           |
| 10              | 4                 | 102                | 202305           | 230.0           |
+-----------------+-------------------+--------------------+------------------+-----------------+
10 rows selected (4.495 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM dim_client
. . . . . . . . . . . . . . . .> ;
INFO  : Compiling command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe): SELECT * FROM dim_client
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_client.id_client, type:int, comment:null), FieldSchema(name:dim_client.nom, type:string, comm
ent:null), FieldSchema(name:dim_client.ville, type:string, comment:null), FieldSchema(name:dim_client.pays, type:string, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe); Time taken: 0.193 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe): SELECT * FROM dim_client
INFO  : Completed executing command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe); Time taken: 0.001 seconds
+-----------------------+-----------------+-------------------+------------------+
| dim_client.id_client  | dim_client.nom  | dim_client.ville  | dim_client.pays  |
+-----------------------+-----------------+-------------------+------------------+
| 1                     | Ali             | Casablanca        | Maroc            |
| 2                     | Sanae           | Tanger            | Maroc            |
| 3                     | Youssef         | Fès               | Maroc            |
| 4                     | Fatima          | Rabat             | Maroc            |
| 5                     | Mohamed         | Agadir            | Maroc            |
+-----------------------+-----------------+-------------------+------------------+
5 rows selected (0.264 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM dim_produit;
INFO  : Compiling command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8): SELECT * FROM dim_produit
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_produit.id_produit, type:int, comment:null), FieldSchema(name:dim_produit.nom_produit, type:s
tring, comment:null), FieldSchema(name:dim_produit.categorie, type:string, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8); Time taken: 0.188 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8): SELECT * FROM dim_produit
INFO  : Completed executing command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8); Time taken: 0.001 seconds
+-------------------------+-------------------------+-----------------------+
| dim_produit.id_produit  | dim_produit.nom_produit | dim_produit.categorie |
+-------------------------+-------------------------+-----------------------+
| 101                     | Laptop Lenovo           | Informatique          |
| 102                     | Smartphone Samsung      | Téléphonie            |
| 103                     | Clavier Logitech        | Accessoires           |
| 104                     | Écran Dell              | Informatique          |
+-------------------------+-------------------------+-----------------------+
4 rows selected (0.227 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM dim_temps;
INFO  : Compiling command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9): SELECT * FROM dim_temps
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_temps.id_temps, type:int, comment:null), FieldSchema(name:dim_temps.jour, type:int, comment:n
ull), FieldSchema(name:dim_temps.mois, type:string, comment:null), FieldSchema(name:dim_temps.annee, type:int, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9); Time taken: 0.111 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9): SELECT * FROM dim_temps
INFO  : Completed executing command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9); Time taken: 0.0 seconds
+---------------------+-----------------+-----------------+------------------+
| dim_temps.id_temps  | dim_temps.jour  | dim_temps.mois  | dim_temps.annee  |
+---------------------+-----------------+-----------------+------------------+
| 202301              | 1               | Janvier         | 2023             |
| 202302              | 5               | Février         | 2023             |
| 202303              | 12              | Mars            | 2023             |
| 202304              | 20              | Avril           | 2023             |
| 202305              | 28              | Mai             | 2023             |
+---------------------+-----------------+-----------------+------------------+
5 rows selected (0.13 seconds)
0: jdbc:hive2://localhost:10000>
```

Partie 4 – Requêtes OLAP

```
0: jdbc:hive2://localhost:10000> SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_temps = t.id_temps GROUP BY t.annee;
INFO  : Compiling command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece): SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_t
emps = t.id_temps GROUP BY t.annee
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:t.annee, type:int, comment:null), FieldSchema(name:_c1, type:double, comment:null)], properties:n
ull)
INFO  : Completed compiling command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece); Time taken: 1.073 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece): SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_t
emps = t.id_temps GROUP BY t.annee
INFO  : Query ID = hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece
INFO  : Tez session hasn't been created yet. Opening session
INFO  : Dag name: SELECT t.annee, SUM(v...... GROUP BY t.annee (Stage-1)
INFO  : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO  : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece], Dag ID: [dag_1774800394520_0001_1], DAG Session ID: [applica
tion_1774800394520_0001]
INFO  : Status: Running (Executing on YARN cluster with App id application_1774800394520_0001)

----------------------------------------------------------------------------------------------
        VERTICES      MODE        STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
----------------------------------------------------------------------------------------------
Map 3 .......... container     SUCCEEDED      1          1        0        0       0       0
Map 1 .......... container     SUCCEEDED      1          1        0        0       0       0
Reducer 2 ...... container     SUCCEEDED      1          1        0        0       0       0
----------------------------------------------------------------------------------------------
VERTICES: 03/03  [==========================>>] 100%  ELAPSED TIME: 2.08 s
----------------------------------------------------------------------------------------------
INFO  : Completed executing command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece); Time taken: 5.682 seconds
+----------+---------------------+
| t.annee  |         _c1         |
+----------+---------------------+
| 2023     | 1804.4000091552734  |
+----------+---------------------+
1 row selected (6.79 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_produit p ON v.id_produit = p.id_produit GROUP BY p.nom_produi
t ORDER BY total DESC LIMIT 3;
INFO  : Compiling command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393): SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_pr
oduit p ON v.id_produit = p.id_produit GROUP BY p.nom_produit ORDER BY total DESC LIMIT 3
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:p.nom_produit, type:string, comment:null), FieldSchema(name:total, type:double, comment:null)], p
roperties:null)
INFO  : Completed compiling command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393); Time taken: 0.356 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393): SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_pr
oduit p ON v.id_produit = p.id_produit GROUP BY p.nom_produit ORDER BY total DESC LIMIT 3
INFO  : Query ID = hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393
INFO  : Session is already open
INFO  : Dag name: SELECT p.nom_produit,......otal DESC LIMIT 3 (Stage-1)
INFO  : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO  : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393], Dag ID: [dag_1774800394520_0001_2], DAG Session ID: [applica
tion_1774800394520_0001]
INFO  : Status: Running (Executing on YARN cluster with App id application_1774800394520_0001)

----------------------------------------------------------------------------------------------
        VERTICES      MODE        STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
----------------------------------------------------------------------------------------------
Map 4 .......... container     SUCCEEDED      1          1        0        0       0       0
Map 1 .......... container     SUCCEEDED      1          1        0        0       0       0
Reducer 2 ...... container     SUCCEEDED      1          1        0        0       0       0
Reducer 3 ...... container     SUCCEEDED      1          1        0        0       0       0
----------------------------------------------------------------------------------------------
VERTICES: 04/04  [==========================>>] 100%  ELAPSED TIME: 1.23 s
----------------------------------------------------------------------------------------------
INFO  : Completed executing command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393); Time taken: 1.325 seconds
+---------------------+---------------------+
|    p.nom_produit    |        total        |
+---------------------+---------------------+
| Smartphone Samsung  | 640.8999938964844   |
| Laptop Lenovo       | 516.1000061035156   |
| Écran Dell          | 341.0500030517578   |
+---------------------+---------------------+
3 rows selected (1.698 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT c.pays, COUNT(*)  FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays;
INFO  : Compiling command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54): SELECT c.pays, COUNT(*)  FROM ventes v JOIN dim_client c ON v.id_client
 = c.id_client GROUP BY c.pays
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:c.pays, type:string, comment:null), FieldSchema(name:_c1, type:bigint, comment:null)], properties
:null)
INFO  : Completed compiling command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54); Time taken: 0.379 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54): SELECT c.pays, COUNT(*)  FROM ventes v JOIN dim_client c ON v.id_client
 = c.id_client GROUP BY c.pays
INFO  : Query ID = hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54
INFO  : Session is already open
INFO  : Dag name: SELECT c.pays, COUNT(......t GROUP BY c.pays (Stage-1)
INFO  : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO  : Tez session was closed. Reopening...
INFO  : Session re-established.
INFO  : Session re-established.
INFO  : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54], Dag ID: [dag_1774801154118_0001_1], DAG Session ID: [applica
tion_1774801154118_0001]
INFO  : Status: Running (Executing on YARN cluster with App id application_1774801154118_0001)

----------------------------------------------------------------------------------------------
        VERTICES      MODE        STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
----------------------------------------------------------------------------------------------
Map 3 .......... container     SUCCEEDED      1          1        0        0       0       0
Map 1 .......... container     SUCCEEDED      1          1        0        0       0       0
Reducer 2 ...... container     SUCCEEDED      1          1        0        0       0       0
----------------------------------------------------------------------------------------------
VERTICES: 03/03  [==========================>>] 100%  ELAPSED TIME: 1.13 s
----------------------------------------------------------------------------------------------
INFO  : Completed executing command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54); Time taken: 1.42 seconds
+---------+------+
| c.pays  | _c1  |
+---------+------+
| Maroc   | 10   |
+---------+------+
1 row selected (1.844 seconds)
0: jdbc:hive2://localhost:10000>
```

Partie 5 : Visualisation des requêtes Hive avec Superset – Étapes détaillées

Étape 1 – Ajouter le service Superset dans Docker

```yaml
  hiveserver2:
    image: apache/hive:${HIVE_VERSION}
    depends_on:
      - metastore
    restart: unless-stopped
    container_name: hiveserver2
    environment:
      HIVE_SERVER2_THRIFT_PORT: 10000
      SERVICE_OPTS: '-Xmx1G -Dhive.metastore.uris=thrift://metastore:9083'
      IS_RESUME: 'true'
      SERVICE_NAME: 'hiveserver2'
    ports:
      - '10000:10000'
      - '10002:10002'
    volumes:
      - warehouse:/opt/hive/data/warehouse
    networks:
      - hive
  superset:
    image: apache/superset
    container_name: superset
    restart: unless-stopped
    ports:
      - "8088:8088"
    environment:
      SUPERSET_SECRET_KEY: 'thisISaSECRET_key'
      ADMIN_USERNAME: admin
      ADMIN_PASSWORD: admin
      ADMIN_FIRST_NAME: Admin
      ADMIN_LAST_NAME: User
      ADMIN_EMAIL: admin@example.com
    volumes:
      - superset_home:/app/superset_home
    depends_on:
      - hiveserver2
    command: >
      /bin/sh -c "
        superset db upgrade &&
        superset fab create-admin --username admin --firstname Admin --lastname User --email admin@example.com --password admin &&
        superset init &&
        superset run -h 0.0.0.0 -p 8088
      "
    networks:
      - hive

volumes:
  hive-db:
  warehouse:
  superset_home:

networks:
  hive:
    name: hive
```

Étape 2 – Installer les outils nécessaires dans le conteneur Superset

1. Ouvrir un terminal root dans le conteneur Superset :

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it superset bash
root@e0bbbd98719b:/app#
```

2. Installer les outils de compilation nécessaires pour les paquets Python natifs :

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it superset bash
root@e0bbbd98719b:/app# apt-get update && apt-get install -y build-essential g++
Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [294 kB]
Fetched 9347 kB in 6s (1660 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu bzip2 cpp cpp-12 dirmngr dpkg-dev fakeroot g++-12 gcc gcc-12 gnupg gnupg-l10n gnupg-utils gpg gpg-agent
  gpg-wks-client gpg-wks-server gpgconf gpgsm libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan8 libassuan0 libatomic1
  libbinutils libcc1-0 libctf-nobfd0 libctf0 libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-12-dev libgdbm-compat4 libgomp1 libgprofng0 libisl23
  libitm1 libjansson4 libksba8 liblocale-gettext-perl liblsan0 libmpc3 libmpfr6 libnpth0 libperl5.36 libquadmath0 libstdc++-12-dev libtsan2 libubsan1 make
  patch perl perl-modules-5.36 pinentry-curses xz-utils
Suggested packages:
  binutils-doc bzip2-doc cpp-doc gcc-12-locales cpp-12-doc dbus-user-session libpam-systemd pinentry-gnome3 tor debian-keyring g++-multilib g++-12-multilib
  gcc-12-doc gcc-multilib manpages-dev autoconf automake libtool flex bison gdb gcc-doc gcc-12-multilib parcimonie xloadimage scdaemon sensible-utils git bzr
  libstdc++-12-doc make-doc ed diffutils-doc perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl libtap-harness-archive-perl pinentry-doc
The following NEW packages will be installed:
```

```
Setting up gcc (4:12.2.0-3) ...
Setting up g++ (4:12.2.0-3) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.9) ...
Processing triggers for libc-bin (2.36-9+deb12u13) ...
root@e0bbbd98719b:/app#
```

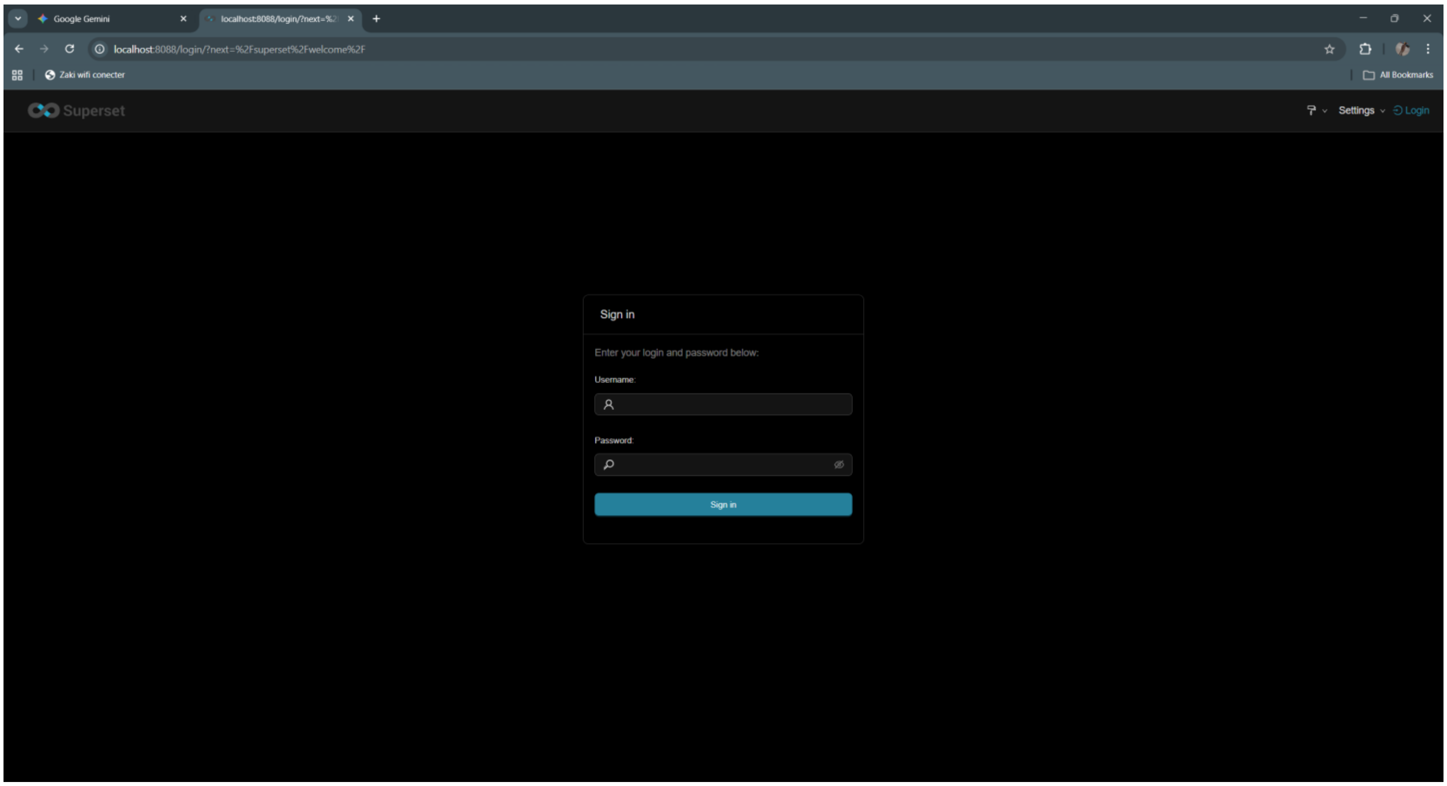
3. Installer les bibliothèques Python permettant à Superset de se connecter à Hive :

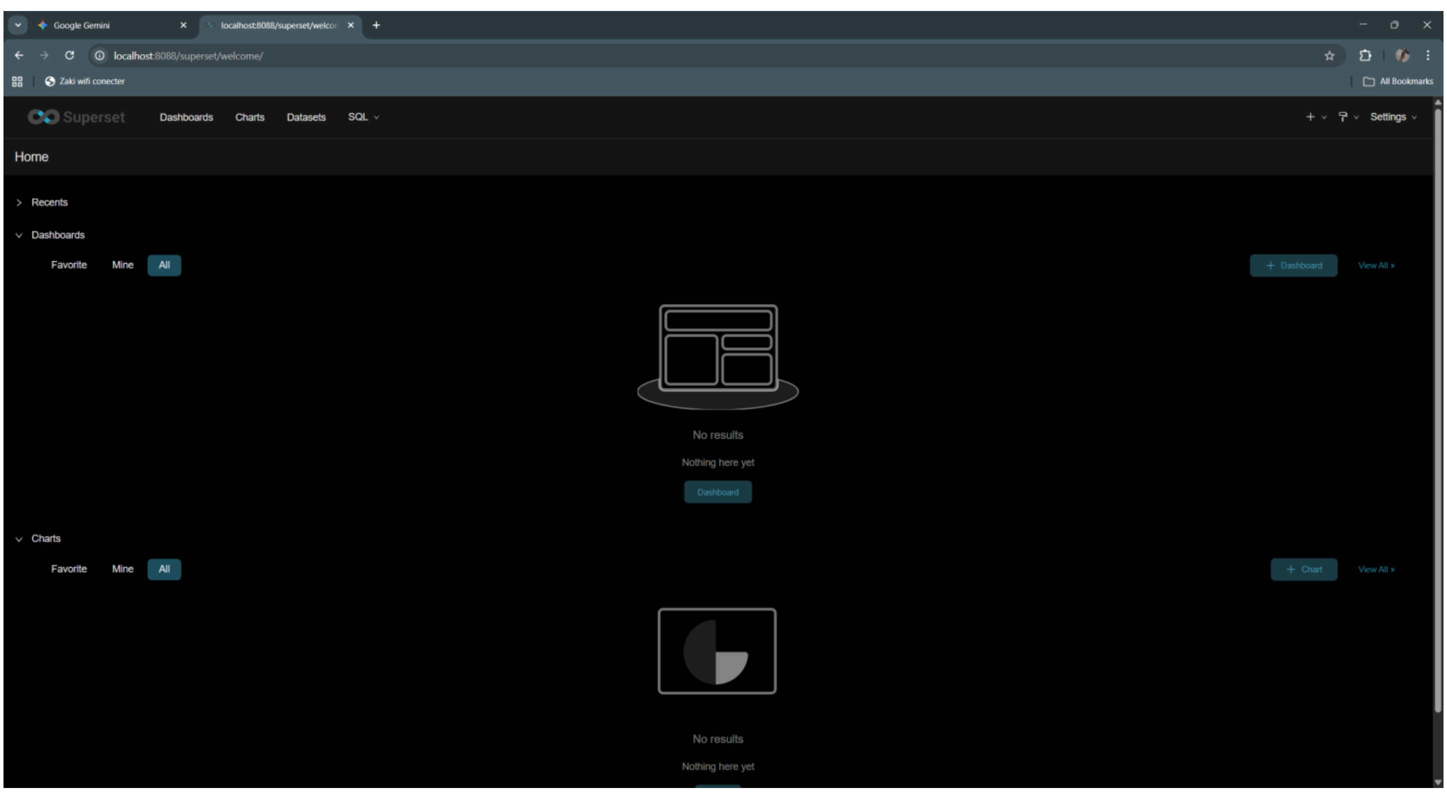
```
root@e0bbbd98719b:/app# pip install "pyhive[hive]" thrift sasl thrift-sasl
WARNING: The directory '/app/superset_home/.cache/pip' or its parent directory is not owned or is not writable by the current user. The cache has been disabled.
Check the permissions and owner of that directory. If executing pip with sudo, you should use sudo's -H flag.
Collecting pyhive[hive]
  Downloading PyHive-0.7.0.tar.gz (46 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.5/46.5 kB 532.5 kB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting thrift
  Downloading thrift-0.22.0.tar.gz (62 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 62.3/62.3 kB 1.6 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting sasl
  Downloading sasl-0.3.1.tar.gz (44 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.7/44.7 kB 2.7 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting thrift-sasl
  Downloading thrift_sasl-0.4.3-py2.py3-none-any.whl (8.3 kB)
Collecting future
```

```
Successfully built thrift sasl pure-sasl pyhive
Installing collected packages: thrift, pure-sasl, six, future, thrift-sasl, sasl, python-dateutil, pyhive
Successfully installed future-1.0.0 pure-sasl-0.6.2 pyhive-0.7.0 python-dateutil-2.9.0.post0 sasl-0.3.1 six-1.17.0 thrift-0.22.0 thrift-sasl-0.4.3
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a vi
rtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
root@e0bbbd98719b:/app#
```

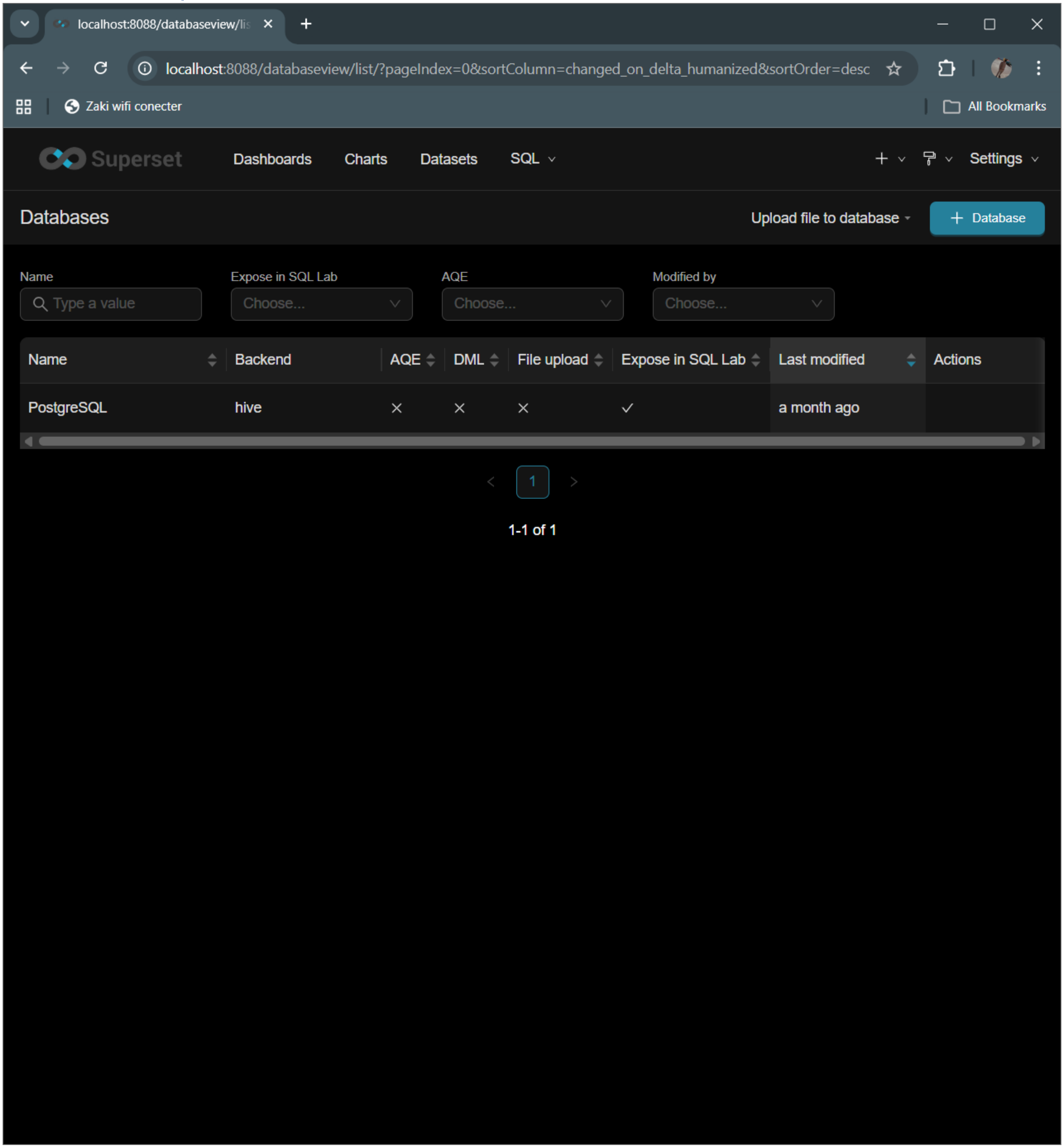
Étape 3 – Démarrer Superset et accéder à l’interface



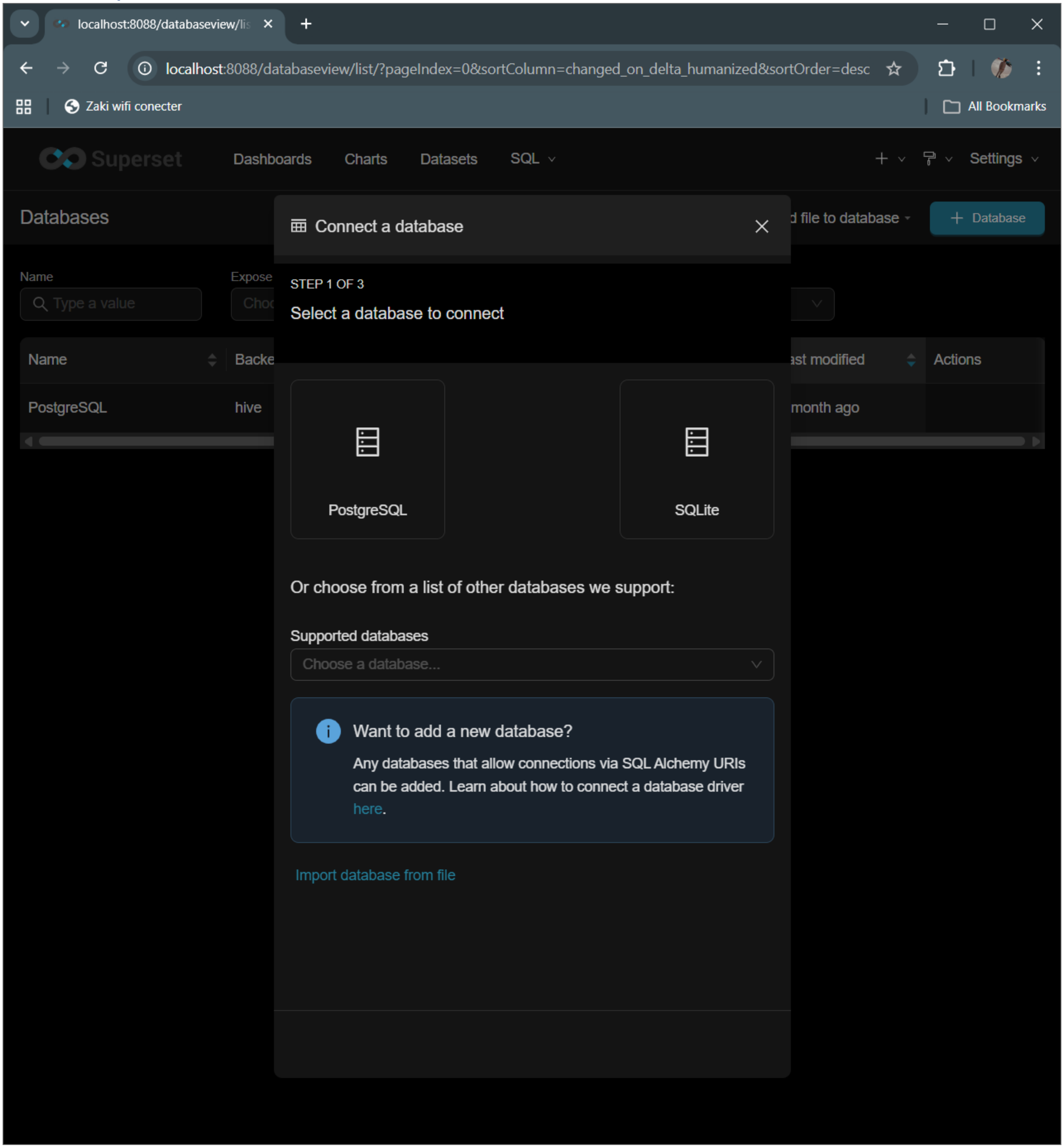


Étape 4 – Connecter Superset à Hive

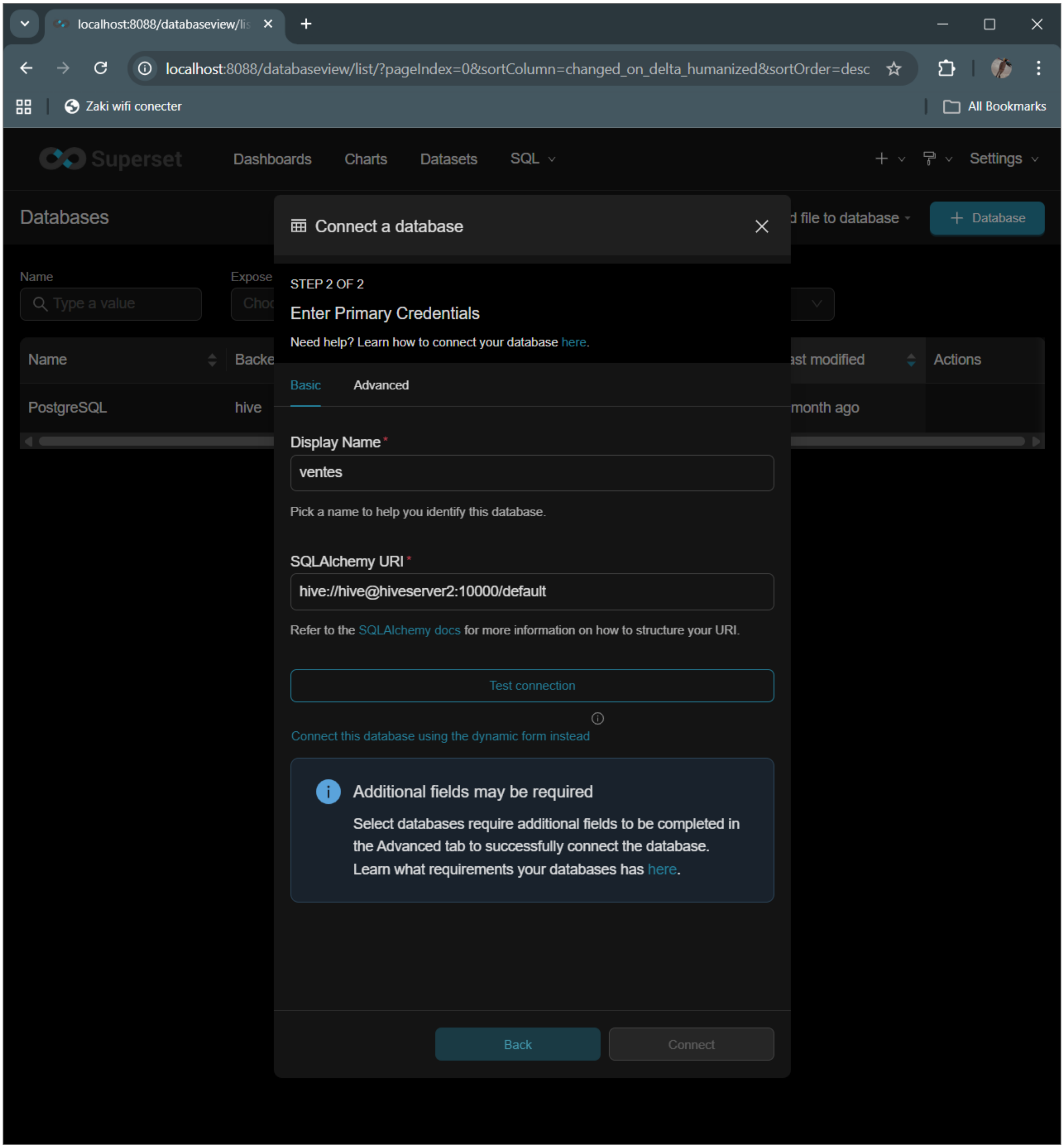
1. *Dans Superset, aller dans le menu Data > Databases*



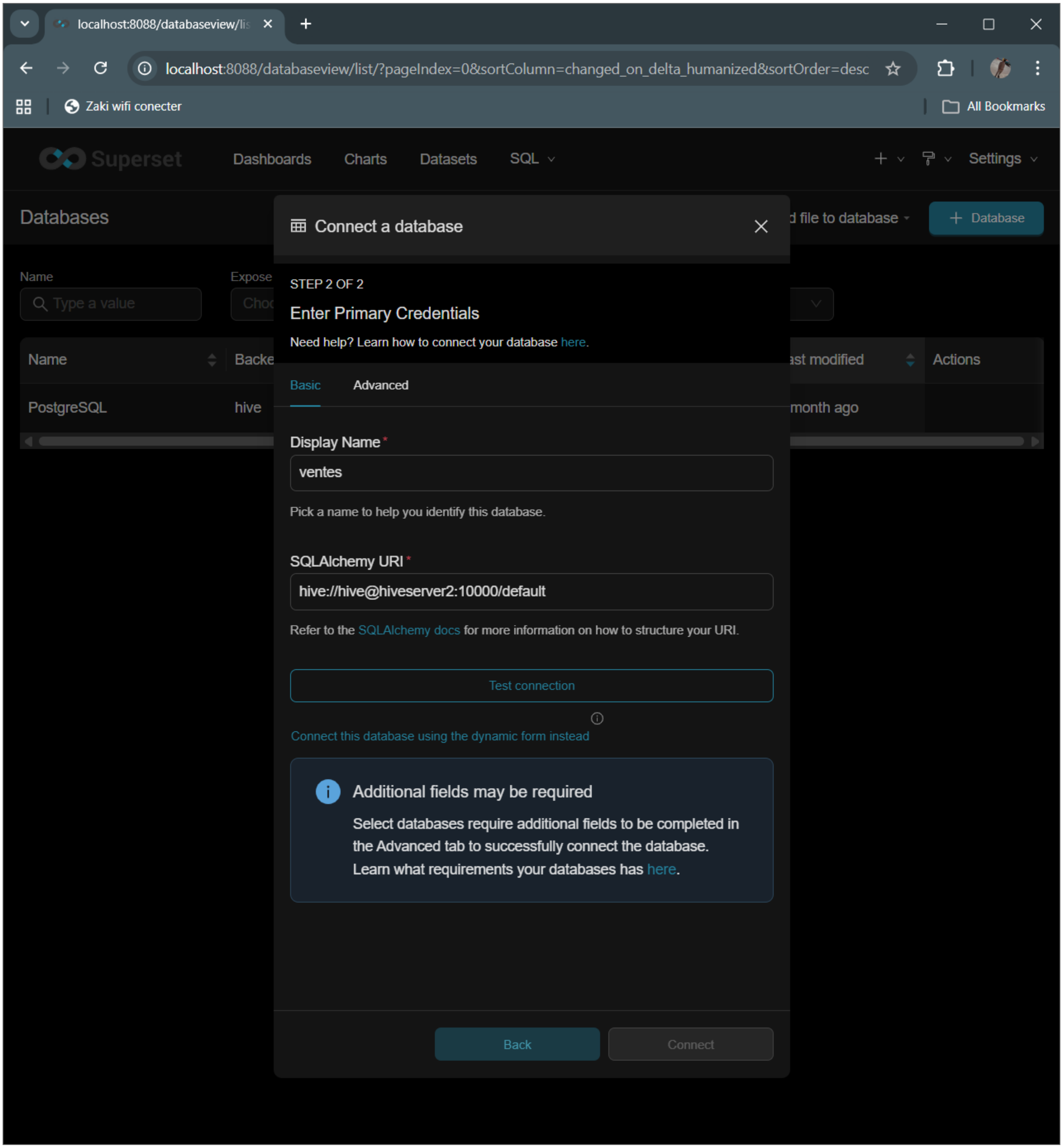
2. Cliquer sur + Database



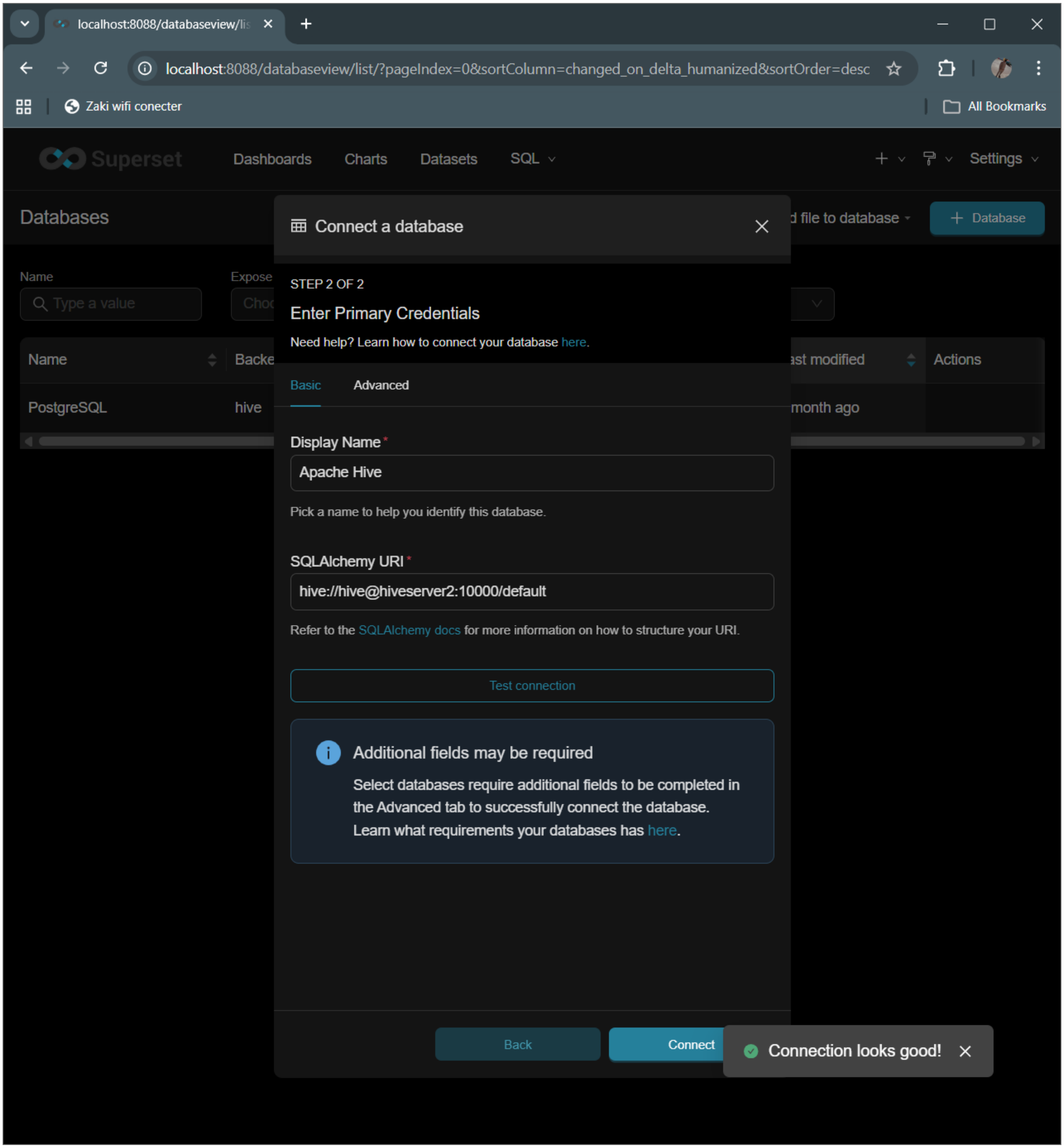
3. Choisir "Connect this database using a SQLAlchemy URI string"



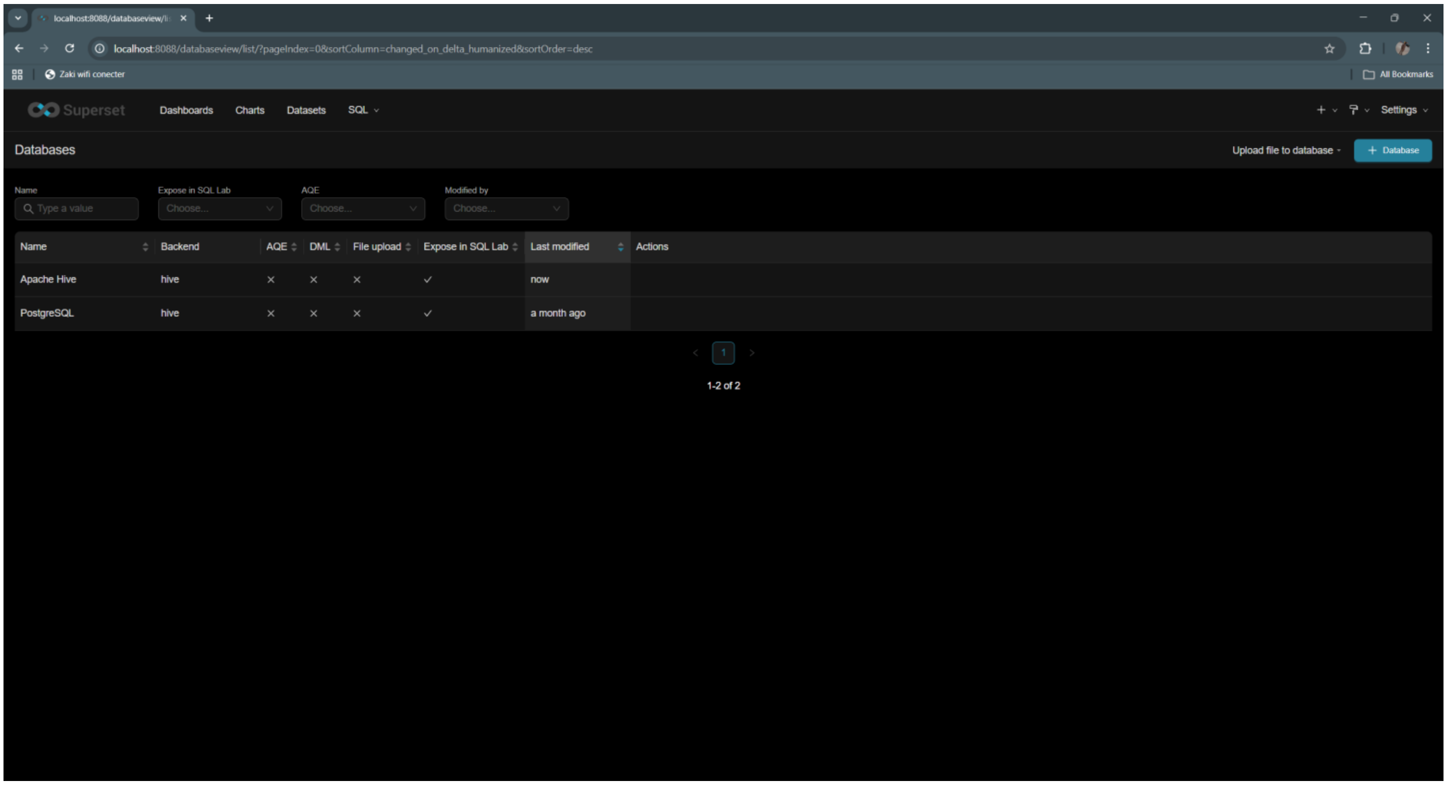
4. Dans le champ `SQLAlchemy URI`, entrer :


localhost:8088/databaseview/li
localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc
Zaki wifi conecter
All Bookmarks
Superset
Dashboards
Charts
Datasets
SQL
Settings
Databases
d file to database
+ Database
Name
Type a value
Expose
Name
Backe
ast modified
Actions
PostgreSQL
hive
month ago
Connect a database
STEP 2 OF 2
Enter Primary Credentials
Need help? Learn how to connect your database here.
Basic
Advanced
Display Name *
ventes
Pick a name to help you identify this database.
SQLAlchemy URI *
hive://hive@hiveserver2:10000/default
Refer to the SQLAlchemy docs for more information on how to structure your URI.
Test connection
Connect this database using the dynamic form instead
Additional fields may be required
Select databases require additional fields to be completed in the Advanced tab to successfully connect the database. Learn what requirements your databases has here.
Back
Connect

5. *Cliquer sur Test Connection*
6. *Si tout est correct, le message* `It looks good!` *s'affiche*

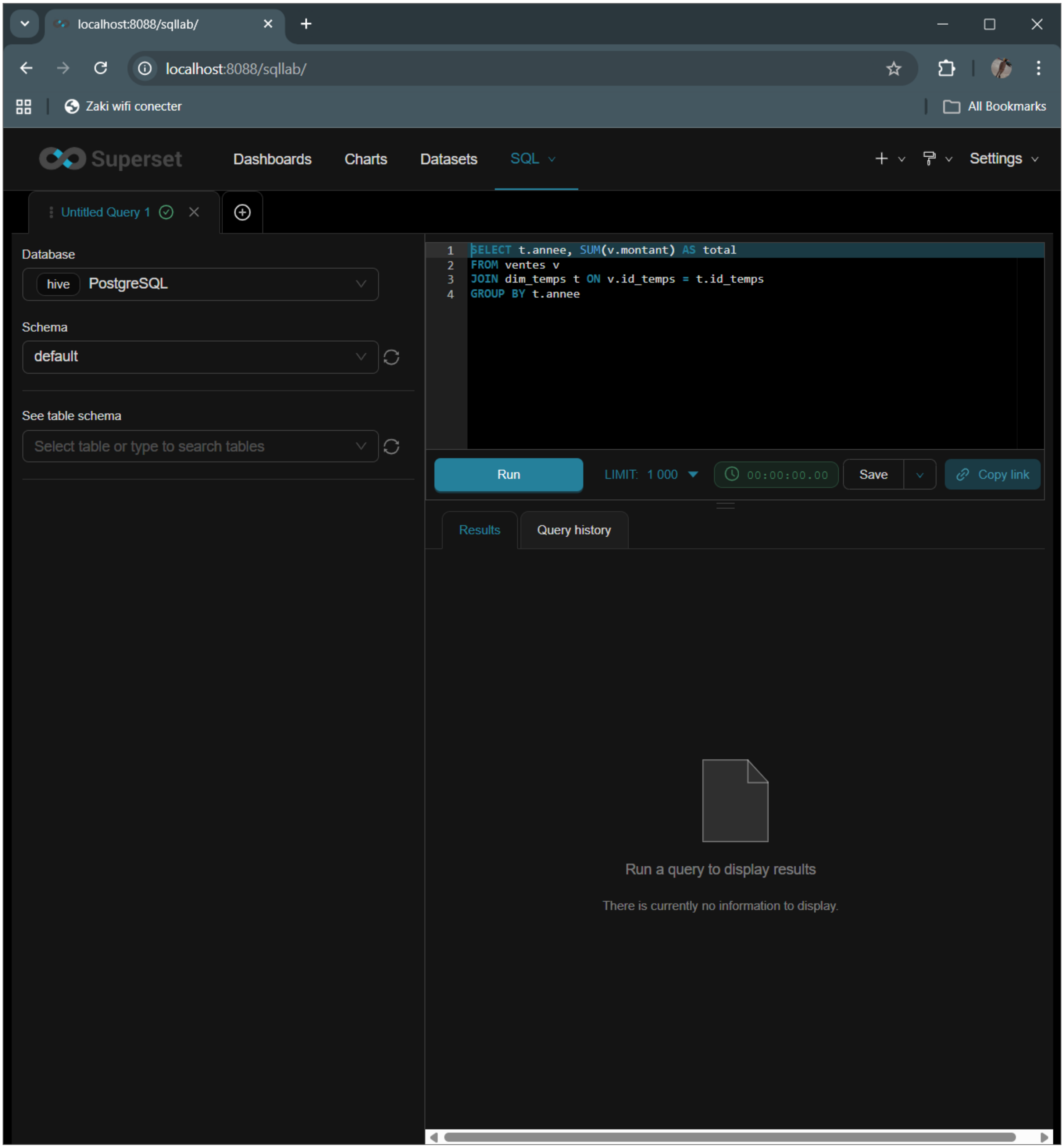


7. *Cliquer sur Save*

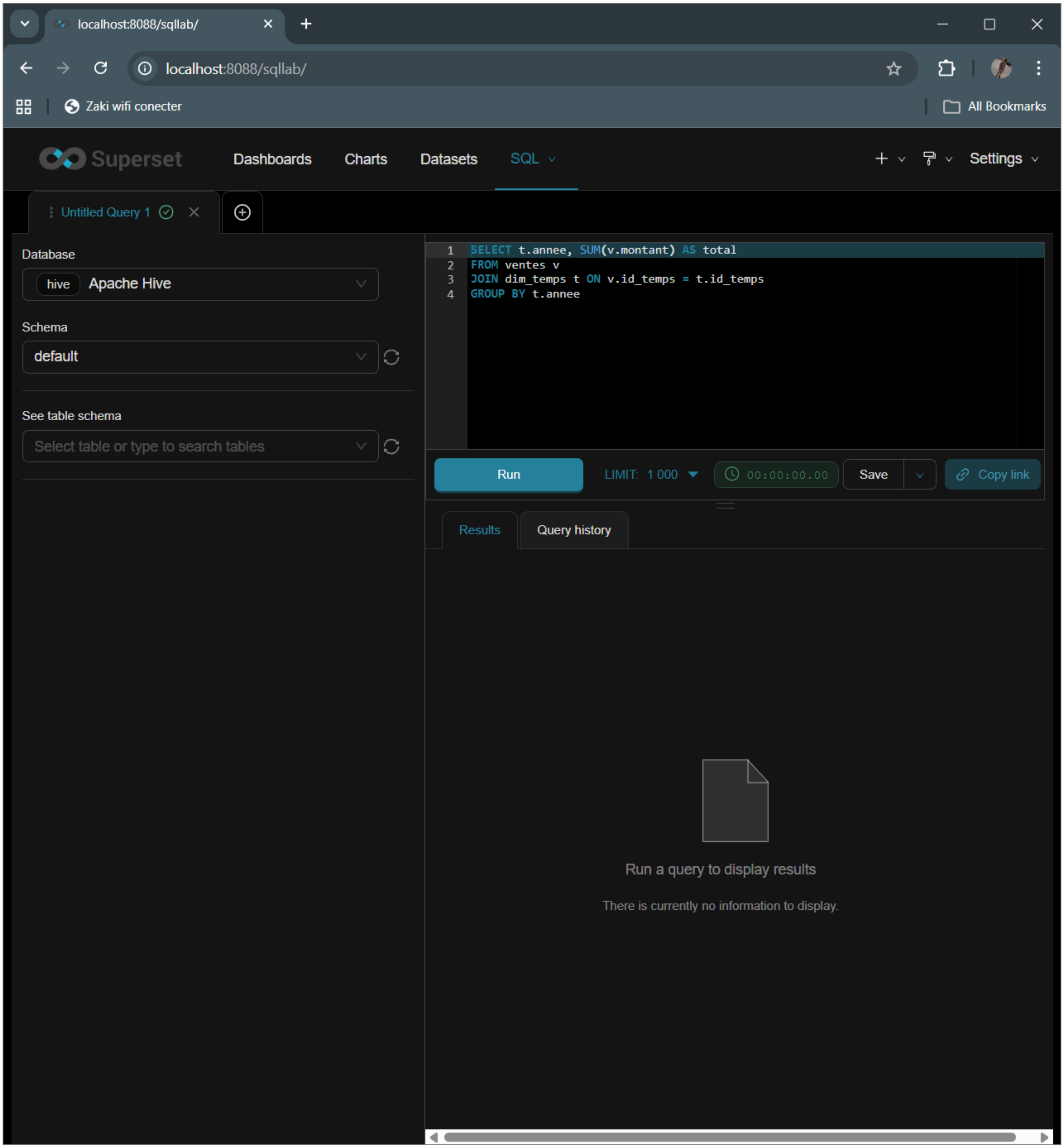

localhost:8088/databaseview/li
localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc
Zaki wifi conecter
All Bookmarks
Superset
Dashboards
Charts
Datasets
SQL
Settings
Databases
Upload file to database
+ Database
Name
Type a value
Expose in SQL Lab
Choose...
AQE
Choose...
Modified by
Choose...
Name
Backend
AQE
DML
File upload
Expose in SQL Lab
Last modified
Actions
Apache Hive
hive
now
PostgreSQL
hive
a month ago
1
1-2 of 2

Étape 5 – Créer une requête Hive

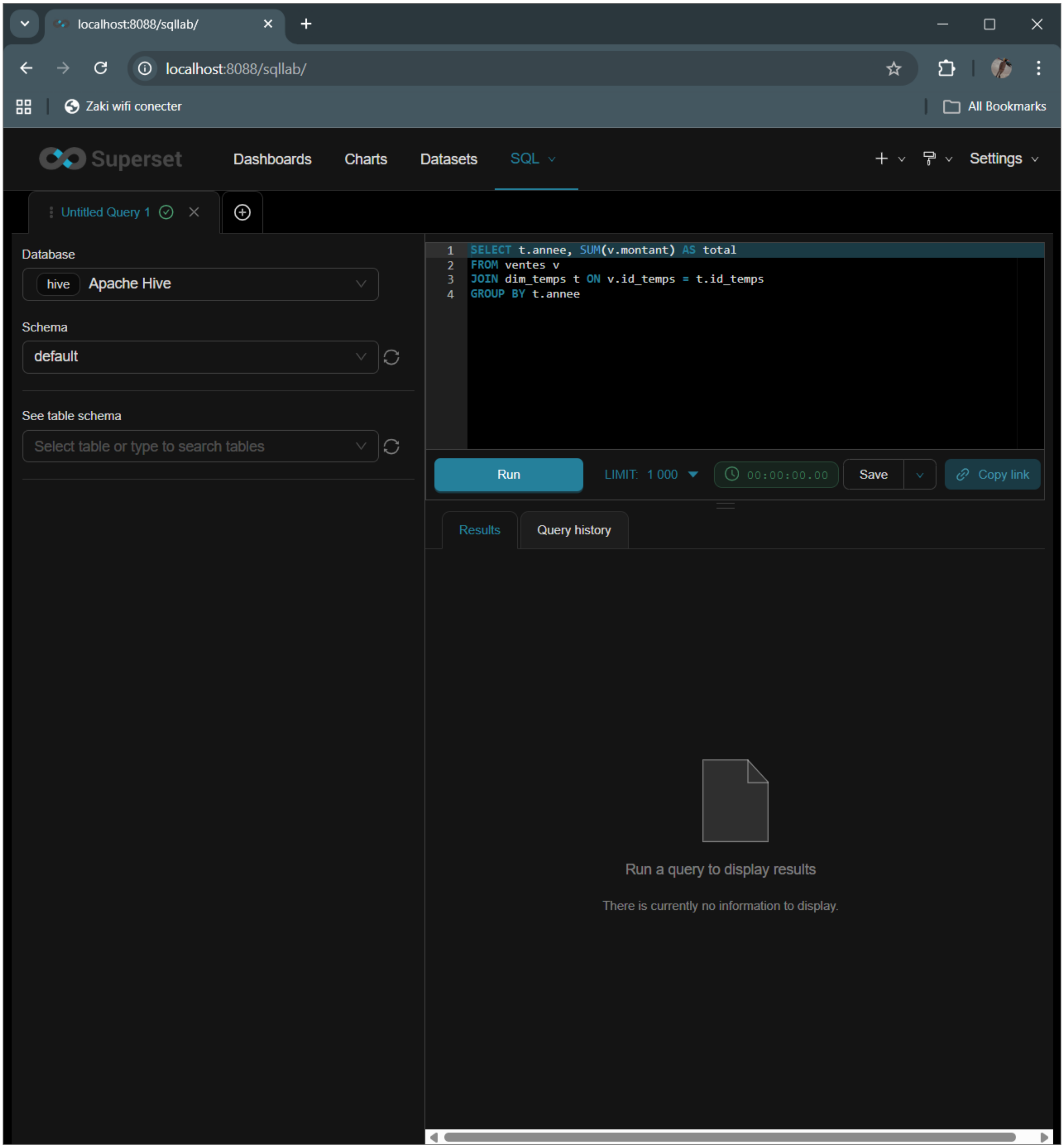
1. *Aller dans SQL Lab > SQL Editor*



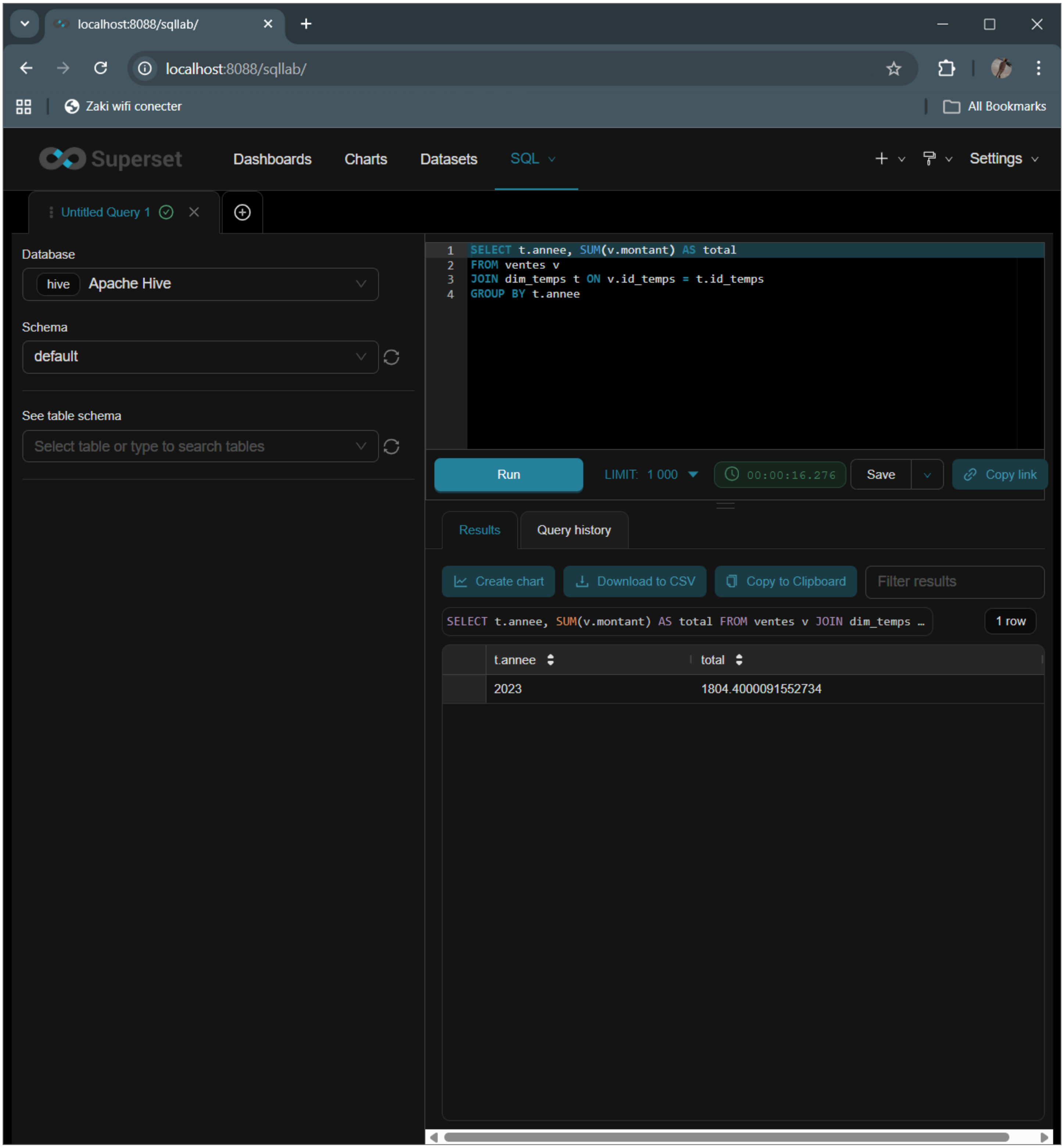
2. *Sélectionner la base de données Hive et le schéma default*



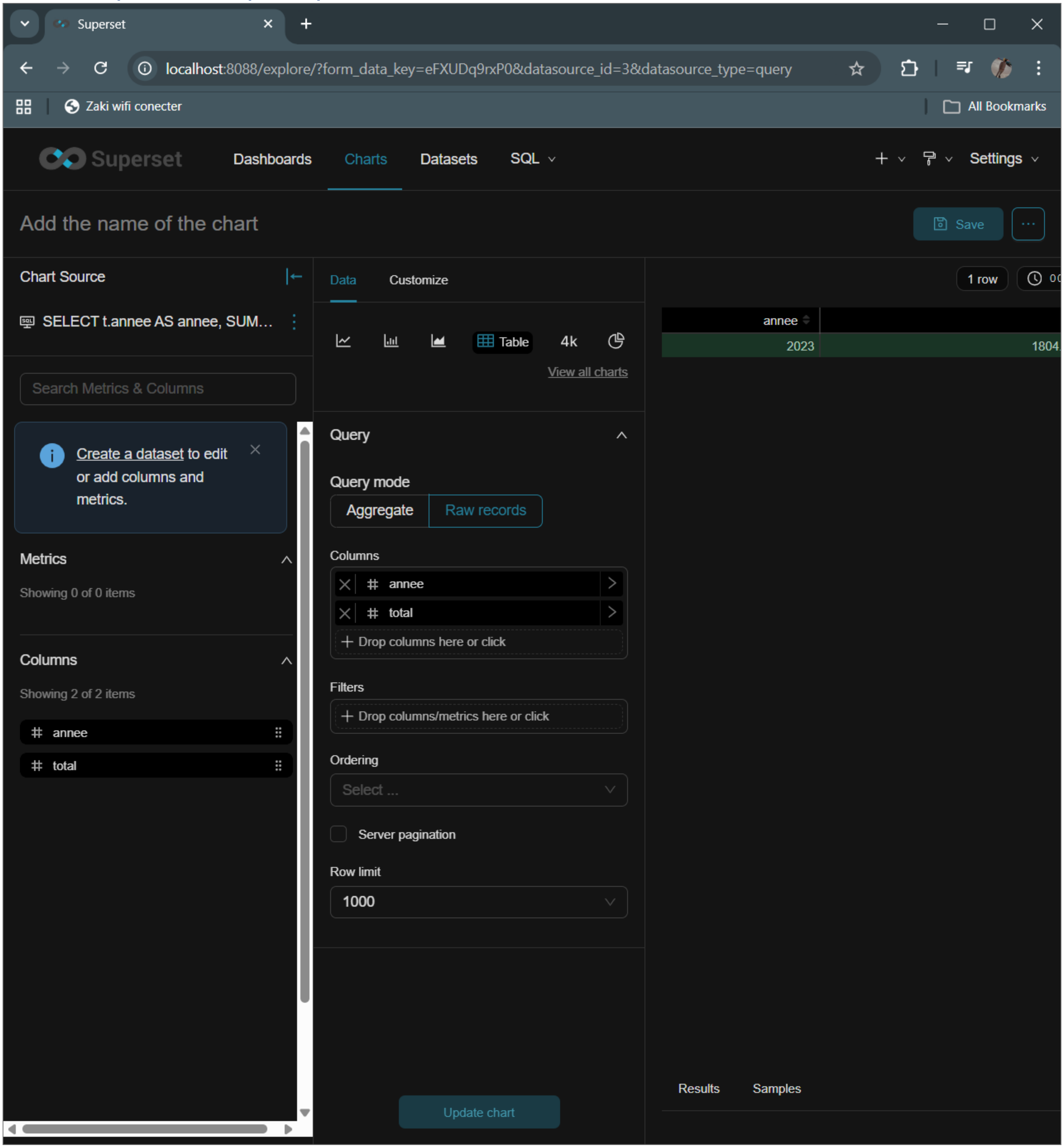
3. Exemple de requête OLAP :



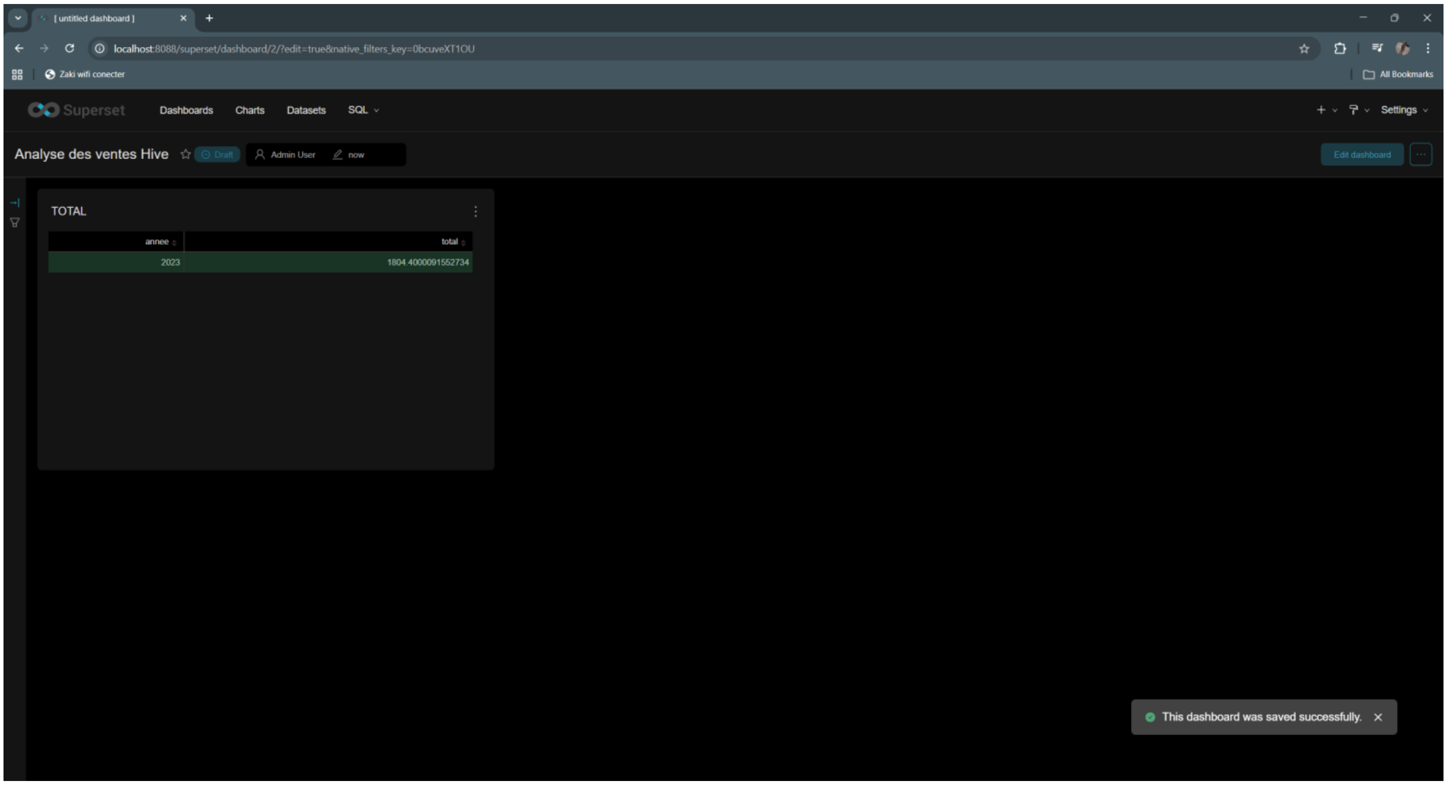
4. Cliquer sur Run pour exécuter la requête



5. *Cliquer sur Explore pour visualiser les résultat*



Étape 6 – Créer un tableau de bord



Partie 6 – Optimisations avancées

```
0: jdbc:hive2://localhost:10000> SET hive.enforce.bucketing = true;
No rows affected (0.04 seconds)
0: jdbc:hive2://localhost:10000> SET hive.enforce.sorting = true;
No rows affected (0.007 seconds)
0: jdbc:hive2://localhost:10000>
```

Partie 6 – Travaux Pratiques

```
0: jdbc:hive2://localhost:10000> -- Création de la tableCREATE TABLE dim_magasin (    id_magasin INT,    nom_magasin STRING,    ville STRING) STORED AS PARQUET;-- Insertion de donnée
s de testINSERT INTO dim_magasin VALUES (10, 'Techno Store Casa', 'Casablanca'),(20, 'Digital Shop Rabat', 'Rabat');
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays;
INFO  : Compiling command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b): SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client G
ROUP BY c.pays
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:c.pays, type:string, comment:null), FieldSchema(name:_c1, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b); Time taken: 0.313 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b): SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client G
ROUP BY c.pays
INFO  : Query ID = hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b
INFO  : Session is already open
INFO  : Dag name: SELECT c.pays, SUM(v.......t GROUP BY c.pays (Stage-1)
INFO  : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO  : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b], Dag ID: [dag_1774822169877_0001_2], DAG Session ID: [application_1774822169877_00
01]
INFO  : Status: Running (Executing on YARN cluster with App id application_1774822169877_0001)

----------------------------------------------------------------------------------------------
        VERTICES      MODE        STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
----------------------------------------------------------------------------------------------
Map 3 .......... container     SUCCEEDED      1          1        0        0       0       0
Map 1 .......... container     SUCCEEDED      1          1        0        0       0       0
Reducer 2 ...... container     SUCCEEDED      1          1        0        0       0       0
----------------------------------------------------------------------------------------------
VERTICES: 03/03  [==========================>>] 100%  ELAPSED TIME: 1.02 s
----------------------------------------------------------------------------------------------
INFO  : Completed executing command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdaea826c5b); Time taken: 1.105 seconds
+---------+---------------------+
| c.pays  |         _c1         |
+---------+---------------------+
| Maroc   | 1804.4000091552734  |
+---------+---------------------+
1 row selected (1.453 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT pays, total_ca FROM ca_par_pays_et_annee;
INFO  : Compiling command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944): SELECT pays, total_ca FROM ca_par_pays_et_annee
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:pays, type:string, comment:null), FieldSchema(name:total_ca, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944); Time taken: 0.214 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944): SELECT pays, total_ca FROM ca_par_pays_et_annee
INFO  : Completed executing command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944); Time taken: 0.0 seconds
+--------+---------------------+
|  pays  |      total_ca       |
+--------+---------------------+
| Maroc  | 1804.4000091552734  |
+--------+---------------------+
1 row selected (0.336 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> DESCRIBE FORMATTED ventes;
INFO  : Compiling command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492): DESCRIBE FORMATTED ventes
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:col_name, type:string, comment:from deserializer), FieldSchema(name:data_type, type:string, comment:from deserializer), FieldSchema(name:comment, type:string, comment:from deserializer)], properties:nu
ll)
INFO  : Completed compiling command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492); Time taken: 0.04 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492): DESCRIBE FORMATTED ventes
INFO  : Starting task [Stage-0:DDL] in serial mode
INFO  : Completed executing command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492); Time taken: 0.086 seconds
+-------------------------------+----------------------------------------------------+----------------------------------------------------+
|           col_name            |                     data_type                      |                      comment                       |
+-------------------------------+----------------------------------------------------+----------------------------------------------------+
| id_vente                      | int                                                |                                                    |
| id_client                     | int                                                |                                                    |
| id_produit                    | int                                                |                                                    |
| id_temps                      | int                                                |                                                    |
| montant                       | float                                              |                                                    |
|                               | NULL                                               | NULL                                               |
| # Detailed Table Information  | NULL                                               | NULL                                               |
| Database:                     | default                                            | NULL                                               |
| OwnerType:                    | USER                                               | NULL                                               |
| Owner:                        | hive                                               | NULL                                               |
| CreateTime:                   | Mon Feb 16 17:51:18 UTC 2026                       | NULL                                               |
| LastAccessTime:               | UNKNOWN                                            | NULL                                               |
| Retention:                    | 0                                                  | NULL                                               |
| Location:                     | file:/opt/hive/data/warehouse/ventes               | NULL                                               |
| Table Type:                   | EXTERNAL_TABLE                                     | NULL                                               |
| Table Parameters:             | NULL                                               | NULL                                               |
|                               | COLUMN_STATS_ACCURATE                              | {\"BASIC_STATS\":\"true\",\"COLUMN_STATS\":{\"id_client\":\"true\",\"id_produit\":\"true\",\"id_temps\":\"true\",\"id_vente\":\"true\",\"montant\":\"true\"}} |
|                               | EXTERNAL                                           | TRUE                                               |
|                               | TRANSLATED_TO_EXTERNAL                             | TRUE                                               |
|                               | bucketing_version                                  | 2                                                  |
|                               | external.table.purge                               | TRUE                                               |
|                               | numFiles                                           | 1                                                  |
|                               | numRows                                            | 10                                                 |
|                               | rawDataSize                                        | 360                                                |
|                               | totalSize                                          | 1330                                               |
|                               | transient_lastDdlTime                              | 1771264437                                         |
|                               | NULL                                               | NULL                                               |
| # Storage Information         | NULL                                               | NULL                                               |
| SerDe Library:                | org.apache.hadoop.hive.ql.io.parquet.serde.ParquetHiveSerDe | NULL                                      |
| InputFormat:                  | org.apache.hadoop.hive.ql.io.parquet.MapredParquetInputFormat | NULL                                    |
| OutputFormat:                 | org.apache.hadoop.hive.ql.io.parquet.MapredParquetOutputFormat | NULL                                   |
| Compressed:                   | No                                                 | NULL                                               |
| Num Buckets:                  | -1                                                 | NULL                                               |
| Bucket Columns:               | []                                                 | NULL                                               |
| Sort Columns:                 | []                                                 | NULL                                               |
| Storage Desc Params:          | NULL                                               | NULL                                               |
|                               | serialization.format                               | 1                                                  |
+-------------------------------+----------------------------------------------------+----------------------------------------------------+
37 rows selected (0.154 seconds)
0: jdbc:hive2://localhost:10000>
```

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes
12K     /opt/hive/data/warehouse/ventes
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes_part
4.0K    /opt/hive/data/warehouse/ventes_part
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes_bucketed
4.0K    /opt/hive/data/warehouse/ventes_bucketed
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive>
```